

Task : Information Gathering

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Objective -

The objective of this assignment was to perform passive and active information gathering on a target system to identify publicly available domain information, discover open ports, identify running services, determine service versions, and understand how reconnaissance is used during the initial phase of a cybersecurity assessment.

Introduction -

Information Gathering (Reconnaissance) is the first phase of every penetration test and security assessment. During this phase, security professionals collect as much information as possible about the target before performing any vulnerability assessment or exploitation.

Reconnaissance is broadly classified into two categories:

- **Passive Reconnaissance** – Information is gathered without directly interacting with the target.
- **Active Reconnaissance** – Information is collected by directly communicating with the target through network requests.

In this assignment, a `whois` lookup was performed as a passive reconnaissance technique, followed by Nmap scans as an active reconnaissance to collect information about the ports and services running on the target machine.

Tools Used :

- Kali Linux – penetration Testing operating system

- whois – retrieving domain registration information
- nmap – network discovery and service enumeration
- Tryhackme target vm – target machine for reconnaissance

Methodology :

Step 1: whois Lookup

The command I ran to gather information available publicly about the domain :

```
whois tutedude.com
```

Command Explanation

Here , whois is the command which retrieves records from the domain registry and “tutedude.com” is the target domain name.

Purpose

The whois command queries the WHOIS database to retrieve publicly available registration information about a domain.

It provides details such as:

- Registrar
- Creation date
- Expiry date
- Name servers
- Domain status

Observation

The whois output contained the following information:

Creation Date: 2019-11-22

Therefore, the domain registration date is:

22/11/2019

This is considered **passive reconnaissance** because no interaction occurs with the target server itself, information is obtained from domain registration databases.

Step 2: Basic Port Scan

The Command I ran to perform a basic port scan is:

```
nmap 192.168.29.188
```

Command Explanation

Here nmap is the network scanning tool and 192.168.29.188 is the target IP address.

Purpose

This scan identifies:

- Live Hosts
- Open TCP ports

Results

The scan identified the following open ports:

```
(kali㉿kali)-[~]
$ nmap -p- 192.168.29.188
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-24 11:50 -0400
Nmap scan report for 192.168.29.188
Host is up (0.000085s latency).
Not shown: 65528 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
80/tcp    open  http
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
8080/tcp  open  http-proxy
65534/tcp open  unknown
MAC Address: 08:00:27:49:75:93 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 2.74 seconds
```

Total Open Ports: 7

Step 3: Service Detection

The command I ran to get detailed information about each port open is:

```
nmap -sV -sC -A 192.168.29.188
```

Explanation of Scan Options

-sV (Version Detection) : Attempts to determine the exact version of services running on open ports.

-sC (Default Script Scan) : Runs Nmap's default NSE (Nmap Scripting Engine) scripts. These scripts safely gather additional information such as -

- SSH host keys
- FTP configuration
- HTTP titles
- SMB information

-A (Aggressive Scan) : Enables several advanced detection features simultaneously like OS Detection ,Version Detection ,Script Scanning ,Traceroute

Observations

Port 21 :-

Service : FTP

Version : vsftpd 3.0.5

Port 22 :-

Service : SSH

Version : OpenSSH 9.6p1 Ubuntu

Port 80 :-

Service : HTTP

Software : Apache httpd

Step 4: Getting the Flag

As the question stated , I performed an advanced nmap scan on the higher port to get the flag and for that I ran the command :

```
nmap 65534 -sV -sC -A 192.168.29.188
```

Observation

```
Not shown: 65528 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
80/tcp    open  http
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
8080/tcp   open  http-proxy
65534/tcp  open  unknown
MAC Address: 08:00:27:49:75:93 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 2.74 seconds

(kali@kali)-[~]
└─$ nmap -p 65534 -sV -sC -A 192.168.29.288
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-24 11:51 -0400
Failed to resolve "192.168.29.288".
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 1.03 seconds

(kali@kali)-[~]
└─$ nmap -p 65534 -sV -sC -A 192.168.29.188
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-24 11:51 -0400
Nmap scan report for 192.168.29.188
Host is up (0.00073s latency).

PORT      STATE SERVICE VERSION
65534/tcp  open  unknown
|_ fingerprint-strings:
|_   GenericLines, NULL,
|_   _ _ _ _ _ FLAG(HIGH_PORT_RECON_SUCCESS)
|_   1 service unrecognized despite returning data. If you know the service/version, please submit the following fingerprint at https://nmap.org/cgi-bin/submit.cgi?new-service :
SF-Port65534-TCP:V=7.99XI=7ND=7/24KTime=6A638A24XP=x86_64-pc-linux-gnuXr(N
SF:FULL,IE,"FLAG(HIGH_PORT_RECON_SUCCESS)\n")Xr(GenericLines,IE,"FLAG(HIGH_
SF:PORT_RECON_SUCCESS)\n");
MAC Address: 08:00:27:49:75:93 (Oracle VirtualBox virtual NIC)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: general purpose
Running: Linux 4.X|5.X
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5
OS details: Linux 4.15 - 5.19, OpenWrt 21.02 (Linux 5.4)
Network Distance: 1 hop

TRACEROUTE
Hop RTT Address
1 0.73 ms 192.168.29.188

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 14.36 seconds

(kali@kali)-[~]
└─$
```

- Discover open ports
- Detect running services

You are not exploiting anything yet.
You are learning how to observe the network.

Answer the questions below

When was the domain tutedude.com registered? (DD/MM/YYYY)

22/11/2019

✓ Correct Answer 0

How many ports are open on the target machine? (Get the IP address of the ova file and perform rest of the assignments on that only).

7

✓ Correct Answer 0

Which port is commonly used for SSH?

22

✓ Correct Answer

Which service is running on port 80?

http

✓ Correct Answer 0

What is the Service Version running on port 21?

vsftpd 3.0.5

✓ Correct Answer 0

Perform an advanced nmap scan on the target to get the flag from higher ports? What is the flag?

FLAG(HIGH_PORT_RECON_SUCCESS)

✓ Correct Answer 0

Step 5: Screenshot Evidence

```
(kali@kali)-[~]
└─$ whois tutedude.com
Domain Name: TUTEDUDE.COM
Registry Domain ID: 2458272606_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.godaddy.com
Registrar URL: http://www.godaddy.com
Updated Date: 2026-05-10T11:24:15Z
Creation Date: 2019-11-22T13:10:34Z
Registry Expiry Date: 2026-11-22T13:10:34Z
Registrar: GoDaddy.com, LLC
Registrar IANA ID: 146
Registrar Abuse Contact Email: abuse@godaddy.com
Registrar Abuse Contact Phone: 480-624-2505
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientRenewProhibited https://icann.org/epp#clientRenewProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Name Server: CHRIS.NS.CLOUDFLARE.COM
Name Server: ELLY.NS.CLOUDFLARE.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-07-24T09:30:09Z <<<
```

You are learning how to observe the network.

Answer the questions below

When was the domain tutedude.com registered? (DD/MM/YYYY)

✓ Correct Answer 👁️

How many ports are open on the target machine?(Get the IP address of the ova file and perform rest of the assignments on that only).

✓ Correct Answer 👁️

Which port is commonly used for SSH?

✓ Correct Answer 👁️

Which service is running on port 80?

✓ Correct Answer 👁️

What is the Service Version running on port 21?

✓ Correct Answer 👁️

Perform an advanced nmap scan on the target to get the flag from the website? What is the flag?

Answer format: *****

Check 👁️

Task 5 Web App Hacking

Task 6 VAPT Scan

Security Significance of the Findings

WHOIS records provide valuable information about a domain, including registration dates, registrar details, and name servers. Security analysts use this information to understand domain ownership, investigate suspicious domains, and perform threat intelligence. Attackers may also use WHOIS data to identify potential targets or infrastructure.

2. Open Ports

Open ports indicate services that are accessible over the network. Every exposed service increases the attack surface of a system. Identifying unnecessary open ports allows administrators to reduce potential entry points by disabling unused services or applying firewall rules.

3. Service Enumeration

Knowing the exact version of a service enables security professionals to determine whether it is up to date or potentially affected by known vulnerabilities. Maintaining an accurate inventory of service versions is an essential part of vulnerability management.

4. SSH Service

SSH is widely used for secure remote administration. While it encrypts communication, an exposed SSH service should be protected with strong passwords or SSH keys, restricted access, and regular monitoring to reduce the risk of unauthorized access attempts.

5. HTTP Service

An HTTP service often hosts web applications that may become targets for attacks such as SQL Injection, Cross-Site Scripting (XSS), directory traversal, or authentication bypass. Identifying the web server is the first step toward evaluating the security of hosted applications.

6. FTP Service

FTP is commonly used for file transfers but, in its standard form, transmits credentials and data in plaintext. Systems running FTP should be carefully configured, avoid anonymous access unless intentionally required, and preferably use secure alternatives such as SFTP or FTPS for sensitive environments.

Real-World Applications

The techniques performed in this assignment are widely used by cybersecurity professionals during :

- Penetration Testing
- Vulnerability Assessments
- Red Team engagements
- Security Audits
- Asset Discovery
- Network Inventory
- Incident Response
- External Attack Surface Management

Reconnaissance allows defenders to identify exposed assets before attackers do and forms the foundation for subsequent security testing.

Conclusion

This assignment demonstrated the practical application of both passive and active reconnaissance techniques. A WHOIS lookup was used to obtain publicly available information about the target domain, while Nmap scans were performed to identify open ports, running services, and service versions on the target machine. The assessment revealed seven open ports, confirmed that SSH was running on port 22, identified HTTP on port 80, determined that the FTP service was **vsftpd 3.0.5** , and found the flag on port 65534. These findings illustrate how reconnaissance helps build an understanding of a target system's exposed services and provides the essential information required for effective vulnerability assessment and defensive security planning.